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Use of successive ionic layer adsorption and reaction (SILAR) method for amorphous titanium dioxide thin films growth

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Abstract

Use of successive ionic layer adsorption and reaction (SILAR) method was preferred for the growth of amorphous titanium dioxide (TiO_2) thin films at ambient temperature. Further, these films were annealed at 673 K for 2 h in air for structural improvement and characterized for structural, surface morphological, optical and electrical properties. An amorphous structure of TiO_2 was retained even after annealing as confirmed from XRD studies. The spherical grains of relatively large size were compressed after annealing. A red shift in band gap energy and decrease in electrical resistivity were observed due to annealing treatment.
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1. Introduction

Titanium dioxide (TiO_2) is one of the technologically important materials, which has been the subject of extensive studies because of its typical chemical, electrical and optical properties such as wide band gap energy, excellent transparency in the visible light, high refractive index and good insulating properties [1,2]. Titanium dioxide has received an enormous attention because of its potential use in photocatalysis, photovoltaic and gas sensing applications [3,4]. The high oxidation affinity, stability and non-toxicity of TiO_2 have stimulated intensive research on environmental applications like dielectrics in memory cell capacitors [5], semiconducting field effect transistors [6], anti-reflection coatings [7], multi-layer optical coatings [8], optical wave-guides, etc. [9]. An amphiphilic TiO_2 surface induced by UV irradiation, which is expected to be applicable to a windshield and a mirror for vehicles [10,11], has recently found potential interest. The compact TiO_2 layers are found to have many applications such as hole blocking layers in 3-D solar cells and/or contaminant barrier between dye inserted porous TiO_2 and concerning electrolytes [12] in addition to the gate dielectric in metal oxide semiconductor field effect transistor [13], surface for solar energy conversion [14], high efficiency photocatalyst [15], etc. On the other hand, in the field of alternative energy, porous TiO_2 electrode in dye-sensitized solar cell is a hot topic due to its high conversion efficiency [16].

The synthesis TiO_2 films have been reported by many workers using different chemical methods such as chemical bath deposition, electrodeposition, sol–gel, etc. [11,12,17]. A successive ionic layer adsorption and reaction (SILAR) method is one of the chemical methods for making uniform and large area thin films, which is based on immersion of the substrate into separately placed cations and anions. The interesting advantages and details of theoretical aspects of this method are greatly described in the review by Pathan and Lokhande [18]. The SILAR method has been successfully employed for many metal sulfides, selenides, tellurides including oxides. Here, for the first time, we demonstrate synthesis of TiO_2 thin films using SILAR method at ambient temperature conditions. As-deposited films were annealed at 673 K for 2 h to observe the annealing effect. Both, as-deposited and annealed films were characterized for their structural, morphological, optical and electrical properties.

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2. Experimental details

2.1. TiO_2 thin film formation

Titanium oxide films were deposited on glass substrates. It is noteworthy that any substrates can be used where main criteria in the selection is that the selected substrates do not react with the bath precursor solutions. In order to remove the oily substances from surface, substrates were boiled in chromic acid for 10 min and then ultrasonically cleaned with double distilled water for few minutes. In the present case, TiO_2 films have been deposited by alternate immersion of substrates in Ti^{4+} source and double distilled water precursor solutions maintained at 333 K, respectively. Titanium isopropoxide was used as Ti^{4+} source, which was dissolved in concentrated H_2SO_4 and the concentration was adjusted to 0.035 M. The pH of the solution was adjusted at 3–4 by addition of aqueous ammonia solution. The oxygen precursor was the double distilled water maintained at temperature of 333 K. In initial process step, the substrate was immersed in titanium isopropoxide solution for 40 s which will adsorbed $(TiOH)^-$ ions on the substrate. After immersion of this substrate in oxygen precursor, for 20 s, the reaction occurred at the substrate surface to form TiO_2, consists one cycle. The rinsing and adsorption time were varied in such a manner that TiO_2 films get deposited onto the substrates. The films were annealed at 673 K for 2 h and the effect of annealing on structural, optical and electrical properties was studied.

2.2. Techniques used for characterization

Thickness of the TiO_2 film after 90 immersion cycles was determined from cross-sectional SEM and was found to be 140 nm. The crystal structure was determined with a Philips PW-3710 X-ray diffractometer using Cu Kα radiation ($\lambda = 1.5406$ Å). The surface morphology of the film was observed with a Philips XL-30 scanning electron microscopy (SEM). For the SEM analyses, TiO_2 film was coated with a 10 nm gold–palladium (Au–Pd) layer by using polaron scanning electron microscopy (SEM) sputter coating unit E-2500, before taking image. The optical absorption spectrum and band gap energy estimation of as-deposited an annealed TiO_2 film were carried out within wavelength range 300–850 nm using Systronics spectrophotometer-119. To determine the electrical resistivity, point two-probe method was used where ohmic contacts were ensured by silver paste.

3. Results and discussion

3.1. Film formation mechanism

The mechanism of TiO_2 film formation by SILAR method can be explained as follows. Titanium isopropoxide dissolved in H_2SO_4 solution at pH ~ 3–4 (adjusted by ammonia) gives titanium hydroxyl $Ti(OH)_5^-$ ions. When substrate is immersed in this solution, $Ti(OH)_5^-$ ions get adsorbed onto the substrate due to attraction between ions in the solution and surface of the substrate. These forces may be cohesive forces or van der Waals forces or chemical attractive forces. Here, we proposed the reaction mechanism of film formation as,

$$Ti_4{}^+ + 5OH^- \rightarrow Ti(OH)_5{}^- \quad (1)$$

The substrate is then immersed in double distilled water to convert titanium hydroxide into titanium oxide by the reduction of hydrogen ions, which can be represented as,

$$Ti(OH)_5{}^- + H^+ \rightarrow TiO_2 + 3H_2O \quad (2)$$

3.2. Structural studies

Film crystallinity was analyzed using X-ray diffraction. The XRD patterns of as-deposited and annealed TiO_2 films onto the glass substrate are shown in Fig. 1. Practically no difference than amorphous structure is seen in as-deposited and annealed TiO_2 thin films. Zhitomirsky [19,20] and Karuppuchamy et al. [21] have reported the TiO_2 films from acidic baths with H_2O_2 as an oxidizing agent with amorphous structure and also found similar observations for electrochemically deposited TiO_2 films from an acidic bath [12].

3.3. Surface morphological study

Fig. 2(a) shows the SEM micrograph of as-deposited TiO_2 thin film at 50,000×. From SEM image, it is seen that the as-deposited TiO_2 film consists of spherical grains of random sizes. After annealing, the surface is seen to be little changed. It is seen from Fig. 2(b) that the almost all grains have occupied uniform size and the surface is seen to be relatively dense, uniform and compact in nature. For more accurate thickness measurement, here, we have elucidated cross-section of SEM image of Fig. 2(a). Cross-view image as seen in Fig. 2(c) measures 240 nm thickness of as-deposited TiO_2 thin films, which was further used for band gap energy measurement analysis. A similar type of behavior has been observed by Lokhande et al. for chemically deposited TiO_2 films from chemical bath [11]. Formation of such type of surface morphology is desired for TiO_2 films, for its application in 3-D solar cells as barrier layer.

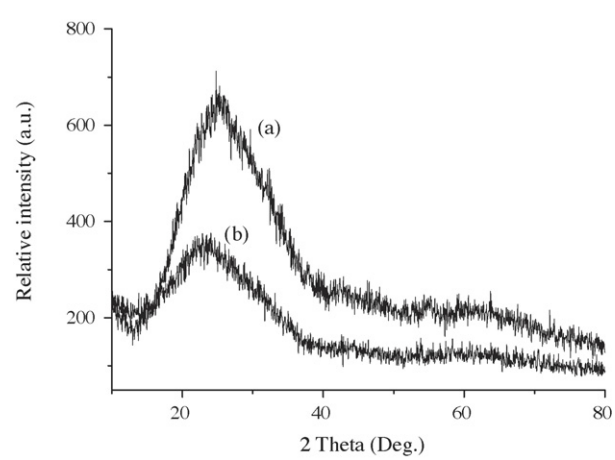

Fig. 1. The XRD patterns of TiO_2 films: (a) as-deposited and (b) annealed at 673 K.

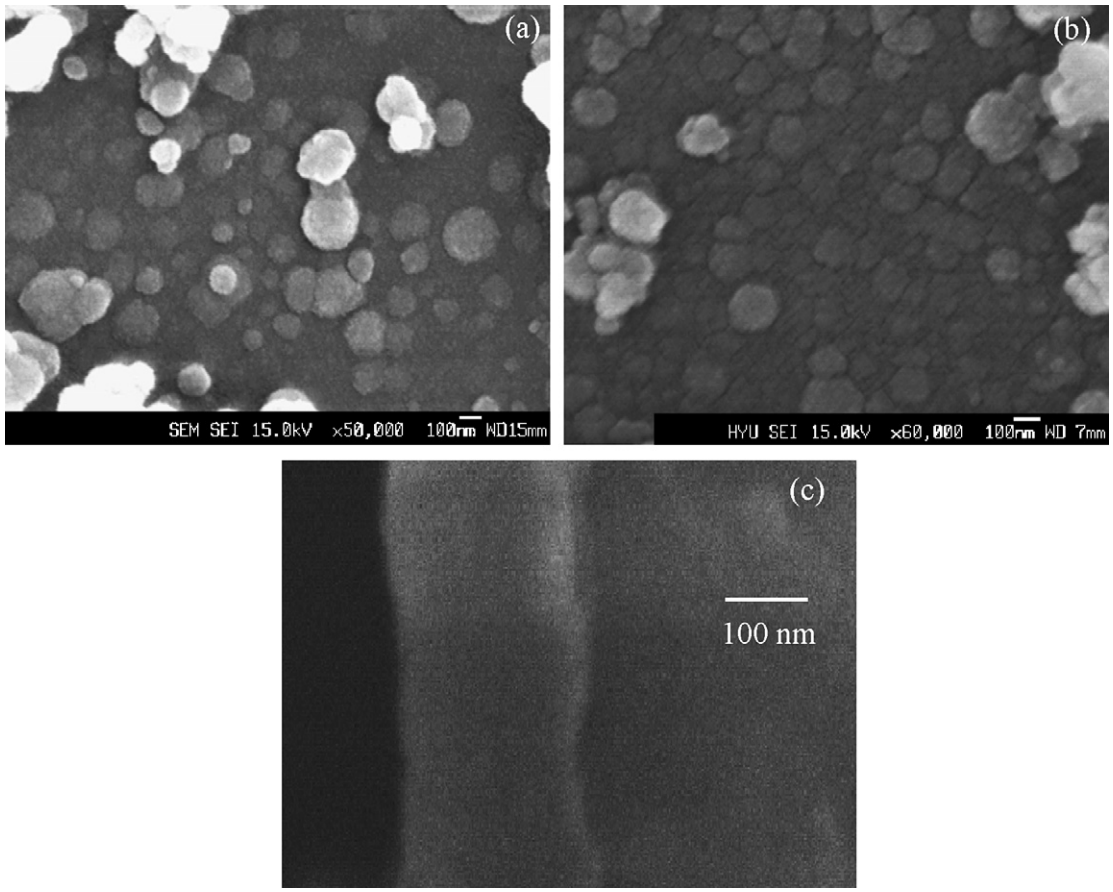

Fig. 2. SEM micrographs of TiO_2 film for: as-deposited (a) and annealed at 673 K (b). Cross-sectional view (c) is depicted for thickness measurement.

3.4. Optical studies

The optical absorption spectra of as-deposited and annealed TiO_2 film are shown in Fig. 3. The absorption edge of the annealed film appeared to shift towards the longer wavelength side. These data were further analyzed from the following classical relation for near edge optical absorption in semiconductors,

$$\alpha = \frac{A(h\nu - E_g)^{n/2}}{h\nu} \tag{3}$$

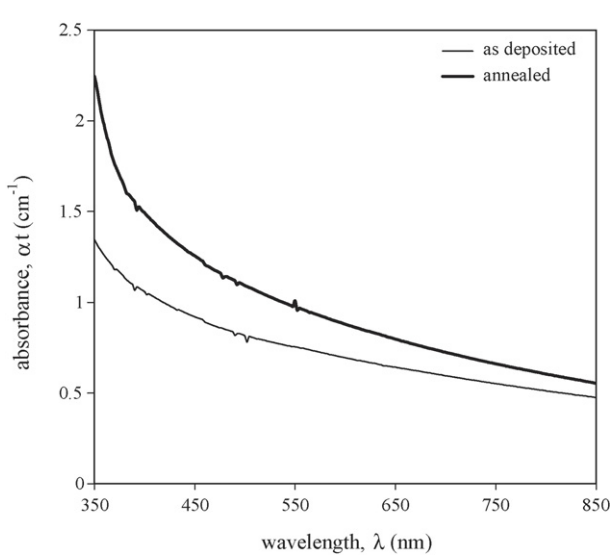

Fig. 3. Optical absorbance of TiO_2 films: as-deposited and annealed at 673 K.

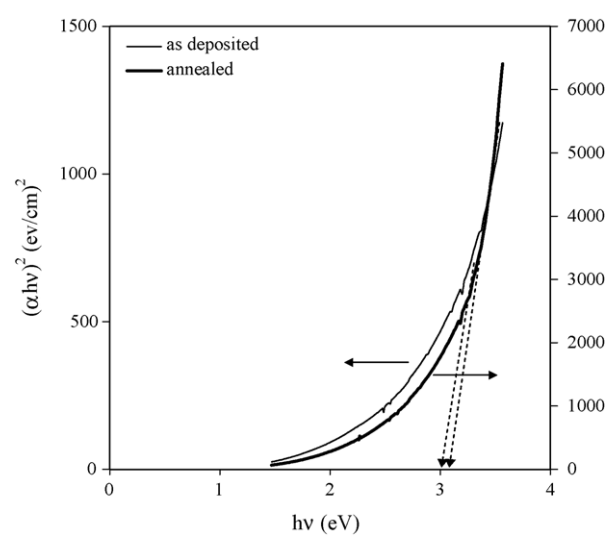

Fig. 4. The plot of $(\alpha h\nu)^2$ against $h\nu$ for the TiO_2 thin films: as-deposited and annealed at 673 K.

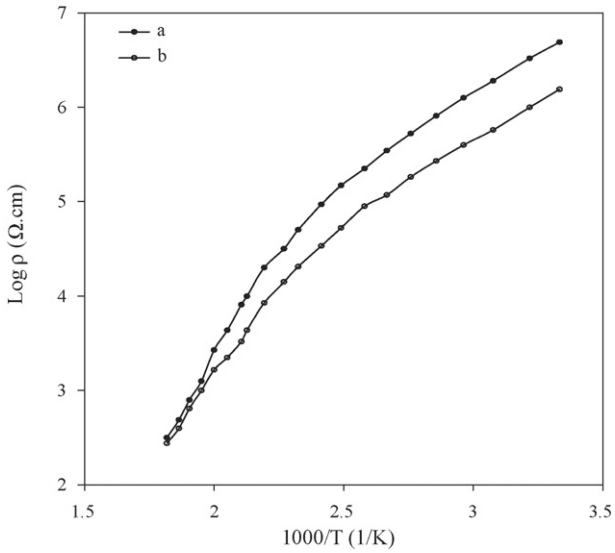

Fig. 5. Variation of dark logarithmic electrical resistivity with reciprocal of temperature for: as-deposited (a) and annealed (b) TiO_2 thin film deposited onto glass substrates.

where A is a constant, E_g the semiconductor band gap energy and n is a number equal to one for direct gap and four for indirect gap compound. The plots of $(\alpha h\nu)^2$ versus $h\nu$ of the as-deposited and annealed films are shown in Fig. 4. Since the plots of $(\alpha h\nu)^2$ verses $h\nu$ are almost linear, the direct nature of the optical transition in TiO_2 is confirmed. Extrapolation of these curves to zero absorption coefficient reveals the optical energy gap as 3.0 eV for annealed TiO_2. The band gap edge of as-deposited TiO_2 seems to be in higher energy region, i.e. UV region of solar spectrum. The higher band gap (3.1 eV) of as-deposited TiO_2 may be due to the inclusion of hydroxide in some extend. However, after annealing the band gap was decreased to 3.0 eV, which is close to band gap of TiO_2. Many researchers have reported this type of behavior of decrease in band gap after annealing [21,22] for chemically deposited metal oxide thin films.

3.5. Resistivity study

The two point-probe dc method was employed to understand the variation of electrical resistivity with temperature in as-deposited and annealed TiO_2 films. The variation of log ρ with reciprocal of temperature (K^{-1}) is depicted in Fig. 5. From figure, it is clearly seen that both as-deposited and annealed TiO_2 films shows negative temperature coefficient of resistance, i.e. resistivity decreases with increase in temperature which is one of the common observations in semiconducting thin films as reported earlier for our compact TiO_2 films [23]. After annealing, the electrical resistivity was decreased by one order of magnitude. This decrease in electrical resistivity of annealed film is may be due to the crystallinity improvement without any structural change which is closely related to phenomena reported elsewhere [24].

4. Conclusions

Here, we have reported the SILAR process for amorphous growth of TiO_2 thin films onto glass substrates at ambient temperature conditions. A morphological study reveals uniform and compact TiO_2 films nature with amorphous structure which remain unchanged even after annealing. Decrease in optical band gap energy and electrical resistivity after annealing is attributed to crystallinity improvement as evidenced from SEM study.

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